TRANSPORT AND TILING BOUNDS FOR THE AFFINE PLANK PROBLEM ON THE TRIANGLE

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Abstract. Bang's affine (relative-width) plank conjecture asserts that if a convex body K is covered by planks $P_1, \ldots, P_n$ then $\sum_i \mathrm{rw}_K(P_i) \geq 1$, where $\mathrm{rw}_K(P) = \mathrm{w}(P)/\mathrm{w}_K(u_P)$ is the width of P relative to the width of K in P's direction. It is open for non-symmetric bodies, already in the plane. We study the triangle, developing a *transport* method (witness measures with uniform marginals) alongside boundary tilings. Main results: (1) three planks parallel to the medians satisfy $\sum \mathrm{rw} \geq 1$, with equality if and only if each is the central third of its median; (2) an explicit witness measure for every cyclic triple whose mid-level lines concur (a two-parameter family), with the bound attained at each member by a covering that is unique for its directions; (3) a characterization: for three pairwise non-parallel directions on the triangle, a witness measure exists if and only if the mid-level lines concur — settling the existence question of Gardner (1988) for three directions on the triangle; (4) a quantification of the method off the concurrent locus by a *transport defect* $D(u)$, with $\sum \mathrm{rw} \geq 1/D(u)$ for coverings by planks parallel to the given triple: $D = \frac{3}{2}$ exactly at the facet triple, and near concurrence $D \leq 1+O(\varepsilon)$, so that planks parallel to any cyclic triple with parameters within $\frac{1}{60}$ of the medians satisfy $\sum \mathrm{rw} \geq \frac{27}{29} > 4\sqrt{3} - 6$ — past the constant of Bakaev–Yehudayoff on the triangle, but only for these restricted direction families, whereas their bound is uniform over all coverings; (5) on the 3-simplex, a one-parameter family of concurrent direction quadruples carrying explicit skeleton witness measures. Supporting results: $\sum \mathrm{rw} \geq 1/d$ for $d \geq 2$ via the uniform measure, sharp two-direction and facet-parallel cases, the "three facets + one plank" case, and sharpness of the chord-lemma step in Bang's sign argument. All our theorems bound coverings by planks in specified direction families on a fixed body, or characterize witness measures; none resolves the conjecture for the triangle, let alone the plane.

1. Introduction

A *plank* of width w is the region between two parallel hyperplanes at distance w. For a convex body $K \subset \mathbb{R}^d$ and a plank P with unit normal u, the *relative width* of P is $\mathrm{rw}_K(P) = \mathrm{w}(P)/\mathrm{w}_K(u)$, where $\mathrm{w}_K(u) = \max_{x\in K}\langle u, x\rangle - \min_{x\in K}\langle u, x\rangle$. Bang's affine plank conjecture [3, 1] states:

Conjecture 1.1 (Affine plank problem; Bang 1951)**.** If K is covered by planks $P_1, \ldots, P_n$, then $\sum_{i=1}^n \mathrm{rw}_K(P_i) \geq 1$.

Ball [2] proved Conjecture 1.1 for centrally symmetric K. For general bodies it is open, and, as emphasised by Bakaev–Yehudayoff [4], it is open *already in the plane* — only the two-plank case is settled there. The best general lower bound is $\sum \mathrm{rw} \geq 2/(1+\sqrt{d})$ [4]; on the equilateral triangle their method gives $\sum \mathrm{rw} \geq 4\sqrt{3} - 6 \approx 0.928$.

Scope. This paper studies the triangle as a concrete body and develops the associated *transport* (witness-measure) and *tiling* methods. It is *not* an attack on Conjecture 1.1 through the Ambrus reduction: that reduction sends a covering by N planks to a simplex of dimension $2N-1$, so

Date: Draft — July 2, 2026.

2020 *Mathematics Subject Classification.* 52C17 (52A40, 52A10).

Key words and phrases. Plank problem, relative width, affine plank conjecture, covering, rigidity, witness measure.

the triangle (the 2-simplex) is never a target (Remark 1.2). Our results are new sharp cases for a fixed body, a rigidity theorem, a dimension-agnostic structural characterization, and an obstruction; we prove nothing about the general conjecture. Since the distinctions matter, we spell them out: five statements should be kept apart — (a) bounds for coverings by planks parallel to a *fixed direction family*; (b) existence and characterization of *witness measures*; (c) bounds for *all* coverings of the triangle; (d) Conjecture 1.1 for the triangle; (e) Conjecture 1.1 in the plane. Every theorem in this paper lives at level (a) or (b); at level (c) we prove nothing beyond the universal $1/d$ of Theorem 2.1, and (d), (e) remain wide open.

Remark 1.2. Ambrus's reduction [1] sends a covering by N planks to a covering of a simplex of dimension $2N-1$ (two points per plank). The target simplices thus have odd dimension growing with N; the 2-simplex (triangle) is never among them. So proving the conjecture for the triangle does *not* imply the planar case, and our results are about the triangle as a concrete body, not a reduction of Conjecture 1.1.

The triangle model. By affine invariance take $\Delta = \{x \in \mathbb{R}^3 : x_i \geq 0,\ x_1 + x_2 + x_3 = 1\}$. A normalized plank is $P = \{x \in \Delta : f(x) \in I\}$ with $f : \Delta \to [0,1]$ affine onto $[0,1]$ and $I \subset [0,1]$ an interval; then $\mathrm{rw}_\Delta(P) = |I|$. Covering means $\Delta \subset \bigcup_i P_i$. The restriction $I \subset [0,1]$ loses no generality: replacing I by $I \cap [0,1]$ leaves $P \cap \Delta$ unchanged and does not increase $|I|$.

Results. Theorem 2.1 (§2), Theorems 3.1 and 3.2 (§3), Theorem 4.1 (§4), and the main results: Theorem 5.1 with rigidity (§5); the weighted-perimeter Theorem 6.3 (§6), which extends the sharp bound from the medians to a two-parameter family of concurrent triples; and Theorem 6.6, which shows the bound is attained at every member of that family by an explicit unique tight covering. Proposition 6.1 (§6) is the dimension-agnostic necessary condition, Corollary 6.4 the resulting characterization for cyclic triples, and Theorem 6.8 the full three-direction characterization on the triangle. Section 7 quantifies the method off the concurrent locus (the transport defect: exact value $\frac{3}{2}$ at the facets, Theorem 7.4; linear stability near concurrence, Theorem 7.6; Corollary 7.7 then gives $\sum \mathrm{rw} \geq \frac{27}{29} > 4\sqrt{3} - 6$ near the medians — for coverings in the given direction families only, see Remark 7.8); §8 constructs the witness family on Δ^3 (Theorem 8.1). Proposition 9.1 and Remark 9.2 (§9) record the normalizer obstruction. Appendix A classifies the boundary tilings of cyclic triples (Proposition A.1), the combinatorial engine behind both rigidity results.

2. The uniform-measure bound $\sum \mathrm{rw} \geq 1/d$

Theorem 2.1. *Let $d \geq 2$. If $K \subset \mathbb{R}^d$ is covered by planks $P_1, \ldots, P_n$, then $\sum_i \mathrm{rw}_K(P_i) \geq 1/d$. The per-plank estimate $\mu(P) \leq d\, \mathrm{rw}_K(P)$ for the uniform measure μ is sharp at the simplex; hence the argument, run with the uniform measure, cannot improve the constant $1/d$. (For $d = 1$ the conjecture itself is trivial: intervals covering an interval satisfy $\sum \mathrm{rw} \geq 1$.)*

Proof. Let μ be the uniform probability measure on K. Fix u, normalize $\langle u, \cdot \rangle \in [0, w]$, $w = \mathrm{w}_K(u)$. The marginal density is $f(t) = V(t)/\operatorname{vol}(K)$ with $V(t) = \operatorname{vol}_{d-1}(K \cap \{\langle u, \cdot \rangle = t\})$; by Brunn–Minkowski $g := V^{1/(d-1)}$ is concave on $[0, w]$. If g peaks at t^* with value g^*, then $g \geq g^* h$ for the tent h ($h(0) = h(w) = 0$, $h(t^*) = 1$), so $f = g^{d-1} \geq (\max f)\, h^{d-1}$ and $1 = \int f \geq (\max f) \int_0^w h^{d-1} = (\max f)\, w/d$, giving $\max f \leq d/w$. Hence $\mu(P) = \int_I f \leq d\, \mathrm{rw}_K(P)$ for each plank, and $1 = \mu(K) \leq \sum_i \mu(P_i) \leq d \sum_i \mathrm{rw}_K(P_i)$. For sharpness of the per-plank estimate: equality in $\max f \leq d/w$ forces $V \propto h^{d-1}$ (conical sections), realized by the simplex $K = \Delta^d$ with u a vertex–facet direction, where a thin plank at the facet has $\mu(P) = (d - o(1))\, \mathrm{rw}_K(P)$. Thus for the simplex the *uniform* measure cannot certify a constant larger than $1/d$ by the union bound; this does *not* assert any covering of the simplex with $\sum \mathrm{rw}$ near $1/d$ (indeed Theorem 3.2 gives $\sum \mathrm{rw} \geq 1$ for facet-parallel coverings). $\square$

Remark 2.2. Other measures do better for *restricted* direction families: the witness measures of §6 certify the full constant 1 for suitable triples, and §7 quantifies the best constant certifiable per triple. For *all* directions simultaneously, however, no single measure certifies the constant 1: uniform marginals in all directions would force $\mathbb{E}[x_k] = \frac{1}{2}$ against $\sum x_k = 1$ (Gardner's obstruction [5]). On the triangle the best constant certifiable by one measure against all directions at once lies between $\frac{1}{2}$ (the uniform measure; Proposition 7.2(iv)) and $\frac{2}{3}$ (the facet triple already forces it; Theorem 7.4); we do not pursue its exact value.

3. Sharp cases: two directions and facet-parallel families

Theorem 3.1. *If every plank covering the triangle is parallel to one of two fixed* non-parallel *directions, then* $\sum \mathrm{rw} \geq 1$*, and this is sharp. (If the two directions are parallel the statement is trivial: all planks share one normalized coordinate* f*, the intervals* $f(P_i)$ *cover* $[0,1]$*, and* $\sum \mathrm{rw} = \sum |I_i| \geq 1$*.)*

Proof. We use Gardner's existence theorem [5, Theorem 1], whose exact statement is: *for a convex body* $K \subset \mathbb{R}^n$ *and two directions* θ_1, θ_2*, there exists a relative-width measure on* K *for* $\{\theta_1, \theta_2\}$ — a probability measure ν, possibly singular, such that $\nu(S \cap K)$ equals the relative width of S for every slab S orthogonal to θ_1 or θ_2; equivalently, the pushforward of ν under each normalized affine coordinate $f_{\theta_i} : K \to [0,1]$ is $\mathrm{Leb}[0,1]$. Applying this to $K = \Delta$ and our two directions gives ν with $f_{u\#}\nu = f_{v\#}\nu = \mathrm{Leb}$. Then $\nu(P) = |I| = \mathrm{rw}(P)$ for each plank, and $1 = \nu(\Delta) \leq \sum \nu(P_i) = \sum \mathrm{rw}(P_i)$. □

Theorem 3.2. *Planks parallel to the facets of a simplex* Δ^d*, in any number, that cover* Δ^d *satisfy* $\sum \mathrm{rw} \geq 1$*, and this is sharp.*

Proof. Write $\Delta^d = \{\lambda \in [0,1]^{d+1} : \sum_k \lambda_k = 1\}$ in barycentric coordinates; a plank parallel to facet k is a coordinate slab $\{\lambda_k \in I\}$. Grouping the planks by facet, let $U_k = \bigcup_{c(i)=k} I_i \subset [0,1]$ be the forbidden set on facet k, so $\sum_k |U_k| \leq \sum_i |I_i| =: R$, and set $C_k = [0,1] \setminus U_k$. A point $\lambda \in \Delta^d$ is uncovered iff $\lambda_k \in C_k$ for every k; since $C_k \subset [0,1]$, an uncovered point exists *iff* $1 \in C_1 + \cdots + C_{d+1}$ (Minkowski sum).

Assume $R < 1$; we produce an uncovered point. Each U_k is a finite union of closed intervals, so C_k is a finite union of intervals, relatively open in $[0,1]$, with $|C_k| = 1 - |U_k| \geq 1 - R > 0$; it need not be compact, so fix $\eta = \frac{1-R}{d+2} > 0$ and choose a *compact* $C'_k \subseteq C_k$ (shave each component) with $|C'_k| \geq |C_k| - \eta$. The one-dimensional Brunn–Minkowski inequality $|A+B| \geq |A| + |B|$ holds for nonempty compact $A, B \subset \mathbb{R}$: $A + B \supseteq (A + \min B) \cup (\max A + B)$, and the two translates meet only at $\max A + \min B$. Iterating it, $D := C'_1 + \cdots + C'_d$ (compact) satisfies $|D| \geq \sum_{k=1}^d |C'_k| \geq d - \sum_{k=1}^d (|U_k| + \eta)$; as $D \subseteq [0,d]$, $|[0,1] \setminus D| \leq |[0,d] \setminus D| \leq \sum_{k=1}^d (|U_k| + \eta)$, whence $|D \cap [0,1]| \geq 1 - \sum_{k=1}^d (|U_k| + \eta)$. Also $(1 - C'_{d+1}) \cap [0,1]$ has measure $|C'_{d+1}| \geq 1 - |U_{d+1}| - \eta$. Both sets lie in $[0,1]$ and their measures sum to at least $2 - R - (d+1)\eta > 1$, so they intersect: there is $t \in D$ with $1 - t \in C'_{d+1}$. Writing $t = \lambda_1 + \cdots + \lambda_d$ with $\lambda_k \in C'_k \subseteq C_k$ and $\lambda_{d+1} = 1 - t \in C_{d+1}$ gives an uncovered $\lambda \in \Delta^d$. Contrapositively, covering forces $R \geq 1$. Sharpness: the facet tiling $I_k = [0, \frac{1}{d+1}]$ gives $\sum \mathrm{rw} = 1$. □

Remark 3.3. The edge-by-edge argument does *not* prove this: on an edge only two barycentric coordinates vary, so no single edge can force $\sum_k |U_k| \geq 1$ across all $d+1$ facets. The Minkowski-sum reduction above is the correct route; note that no witness *measure* can prove it either, since facet directions violate the concurrence condition of Proposition 6.1.

4. Three facets plus one arbitrary plank

Theorem 4.1. *Let $P_i = \{\lambda_i \in I_i\}$ $(i = 1, 2, 3)$ be facet-parallel with $a_i = |I_i|$, and let $Q = \{f \in J\}$, $t = |J|$, with $f : \Delta \to [0, 1]$ affine normalized. If $\Delta \subset P_1 \cup P_2 \cup P_3 \cup Q$ then $a_1 + a_2 + a_3 + t \geq 1$.*

Proof. Choose an active edge of f; relabelling, $f(v_j) = 0$, $f(v_k) = 1$, and let v_i be the opposite vertex, $\theta = f(v_i)$. For $s \in [0, 1]$ consider the fibre $F_s = \{\lambda_i = s\}$, parametrized by $u = \lambda_k \in [0, 1 - s]$, so $\lambda_j = 1 - s - u$ and $f|_{F_s}(u) = \theta s + u$. On F_s: $P_i \cap F_s = F_s$ if $s \in I_i$ (else $\varnothing$); $P_j \cap F_s$, $P_k \cap F_s$, $Q \cap F_s$ have lengths $\leq a_j, a_k, t$. Put $B = a_j + a_k + t$. If $s \notin I_i$ and F_s is covered, its length $1 - s \leq B$. If $B \geq 1$ we are done; else $[0, 1 - B) \subset I_i$, so $a_i \geq 1 - B$. In both cases $a_1 + a_2 + a_3 + t = a_i + B \geq 1$. □

5. The median theorem and its rigidity

Let m_1, m_2, m_3 be the medians of Δ as normalized affine forms: with A, B, C the vertices,

$$m_1 = (\tfrac{1}{2}, 0, 1), \quad m_2 = (0, 1, \tfrac{1}{2}), \quad m_3 = (1, \tfrac{1}{2}, 0),$$

(vertex values). One has $m_1 + m_2 + m_3 \equiv \frac{3}{2}$. A *median plank* is $P_i = \{m_i \in I_i\}$, $I_i = [l_i, h_i]$, $r_i = |I_i| > 0$.

Theorem 5.1. *If three median planks cover Δ, then $r_1 + r_2 + r_3 \geq 1$, with equality if and only if $I_1 = I_2 = I_3 = [\frac{1}{3}, \frac{2}{3}]$.*

We are not aware of a previous tight *non-facet* three-direction rigidity result for the triangle: the sharp cases recorded in the survey [6] (and in [5, 4]) concern facet/coordinate directions, two directions, or symmetric bodies.

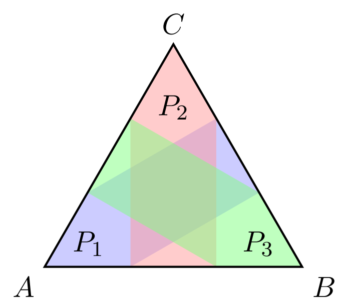

Figure 1. The tight median covering of Theorem 5.1: $I_1 = I_2 = I_3 = [\frac{1}{3}, \frac{2}{3}]$. Each plank $P_i = \{m_i \in [\frac{1}{3}, \frac{2}{3}]\}$ is the central-thirds band of its median; the three traces tile each edge, and the planks cover Δ with $\sum \mathrm{rw} = 1$.

Proof of the bound. Let μ be the perimeter measure: uniform in the parameter on each of the three edges, mass $\frac{1}{3}$ each. A direct computation shows each pushforward $m_{i\#}\mu = \mathrm{Leb}[0, 1]$. Indeed for m_1: on AB, m_1 ranges over $[0, \frac{1}{2}]$ with Jacobian 2 (density $\frac{1}{3} \cdot 2$); on CA, over $[\frac{1}{2}, 1]$ likewise; on BC, over $[0, 1]$ with Jacobian 1 (density $\frac{1}{3}$); the total density is 1 on $[0, 1]$. Hence $\mu(P_i) = \mathrm{Leb}(I_i) = r_i$, and $1 = \mu(\Delta) \leq \sum \mu(P_i) \leq \sum r_i$. □

Proof of rigidity. Equality in $1 \leq \sum \mu(P_i)$ forces the P_i to partition μ a.e.; since μ lives on the edges, *on each edge the three traces tile* $[0, 1]$. We claim there are exactly three such edge-tilings, and only one covers the interior.

Three edge-tilings. On each edge each median is affine, so the three traces are explicit intervals depending piecewise-linearly on (l_i, h_i) through $a_i = |I_i \cap [0, \frac{1}{2}]|$ and $b_i = |I_i \cap [\frac{1}{2}, 1]|$. Classifying each I_i by crossing type (contained in $[0, \frac{1}{2}]$, crossing $\frac{1}{2}$, or contained in $[\frac{1}{2}, 1]$) and each edge by the left-to-right order of its traces, the abutting (tiling) conditions become a finite family of linear systems in (l_i, h_i).

Lemma 5.2. *The edge-tiling conditions have exactly three solutions with* $0 \le l_i < h_i \le 1$*:* $[\frac{1}{3}, \frac{2}{3}]^3$*,* $[0, \frac{1}{3}]^3$*, and* $[\frac{2}{3}, 1]^3$*.*

This is the case $\tau_i = \frac{1}{2}$ of Proposition A.1, proved in full in Appendix A for arbitrary cyclic triples: the symmetries of the configuration reduce the a priori 27 crossing patterns to seven classes, five of which are impossible and two of which lead to cyclic linear systems with unique solutions. The lemma has also been verified by three independent exact computer enumerations, archived as ancillary files.

Interior witness. The centroid G has $m(G) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Since $\frac{1}{2} \notin [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, the tilings $[0, \frac{1}{3}]^3$ and $[\frac{2}{3}, 1]^3$ leave G uncovered; they tile the boundary but are not coverings of Δ. The tiling $[\frac{1}{3}, \frac{2}{3}]^3$ covers Δ and has $\sum r_i = 1$. Hence the unique covering with $\sum r_i = 1$ is $[\frac{1}{3}, \frac{2}{3}]^3$. $\square$

6. A dimension-agnostic necessary condition for transport

The proofs of Theorems 3.1 and 5.1 share one device: a probability measure whose pushforward under each direction is uniform. Which direction-tuples admit one?

Proposition 6.1. *Let* $u_1, \dots, u_{d+1}$ *be normalized affine forms on* Δ^d *with vertex-value matrix* V *invertible. If a probability measure* μ *on* Δ^d *has* $u_{i\#}\mu = \mathrm{Leb}[0,1]$ *for all* i*, then its barycenter* $p \in \Delta^d$ *satisfies* $Vp = \frac{1}{2}\mathbf{1}$*; equivalently the mid-level hyperplanes* $\{u_i = \frac{1}{2}\}$ *concur at* p*, which requires* $\mathbf{1}^\top V^{-1}\mathbf{1} = 2$ *and* $V^{-1}\mathbf{1} \ge 0$*.*

Proof. $\frac{1}{2} = \mathbb{E}_\mu[u_i] = \sum_j V_{ij}\mathbb{E}_\mu[x_j] = (Vp)_i$, and $p \in \Delta^d$ gives $\mathbf{1}^\top p = 1$, $p \ge 0$; substitute $p = \frac{1}{2}V^{-1}\mathbf{1}$. $\square$

The three medians satisfy this with p the centroid; the three facets do not ($\mathbf{1}^\top V^{-1}\mathbf{1} = 3 \ne 2$), recovering Gardner's obstruction. The concurrence family is two-dimensional (parametrized by p): a normalized direction has vertex values $\{0, \tau, 1\}$, and $Vp = \frac{1}{2}\mathbf{1}$ ties each τ_i to p. The medians are the case $p =$ centroid, forcing $\tau_i = \frac{1}{2}$. Among these, the explicit perimeter measure singles out the medians:

Proposition 6.2. *For a direction* u *with vertex values* $\{0, \tau, 1\}$*, the perimeter measure* μ *(uniform on each edge, mass* $\frac{1}{3}$*) satisfies* $u_{\#}\mu = \mathrm{Leb}[0,1]$ *if and only if* $\tau = \frac{1}{2}$*.*

Proof. On the three edges u is affine between the pairs $\{0,\tau\}, \{\tau,1\}, \{0,1\}$, so $u_{\#}\mu$ has density $\frac{1}{3\tau} + \frac{1}{3}$ on $[0,\tau]$ and $\frac{1}{3(1-\tau)} + \frac{1}{3}$ on $[\tau, 1]$. Both equal 1 iff $\tau = \frac{1}{2}$. $\square$

Uniform edge-weights thus single out the medians. Allowing *unequal* edge masses, however, the construction covers a full two-parameter family:

Theorem 6.3 (weighted perimeter)**.** *Let* $p = (p_A, p_B, p_C)$ *be interior with* $\max_j p_j < \frac{1}{2}$*, and let* $u_1 = (\tau_1, 0, 1)$*,* $u_2 = (0, 1, \tau_2)$*,* $u_3 = (1, \tau_3, 0)$ *(vertex values at* A, B, C*) be the cyclic directions concurrent at* p*, i.e.* $\tau_1 = \frac{1/2 - p_C}{p_A}$*,* $\tau_2 = \frac{1/2 - p_B}{p_C}$*,* $\tau_3 = \frac{1/2 - p_A}{p_B}$*. Let* μ_p *be uniform on each edge with masses* $w_{AB} = 1 - 2p_C$*,* $w_{BC} = 1 - 2p_A$*,* $w_{CA} = 1 - 2p_B$ *(summing to* 1*). Then* $u_{i\#}\mu_p = \mathrm{Leb}[0,1]$ *for* $i = 1, 2, 3$*; consequently every covering of* Δ *by planks in these three directions satisfies* $\sum \mathrm{rw} \ge 1$*.*

Proof. Each u_i is affine of full range $[0,1]$ on the edge joining its 0- and 1-vertices (u_1 on BC, u_2 on AB, u_3 on CA) and affine of range $[0, \tau_i]$, resp. $[\tau_i, 1]$, on the other two. The pushforward of a uniform edge mass w_e through an affine map of range length L is uniform with density

w_e/L. Hence $u_{1\#}\mu_p$ has density $w_{BC} + w_{AB}/\tau_1$ on $[0, \tau_1]$ and $w_{BC} + w_{CA}/(1-\tau_1)$ on $[\tau_1, 1]$, and similarly for u_2, u_3: six equations. Using $p_A + p_B + p_C = 1$, each composite term collapses by the identity $(1-2x)\,p/(\frac{1}{2}-x) = 2p$: e.g. $w_{AB}/\tau_1 = (1-2p_C)p_A/(\frac{1}{2}-p_C) = 2p_A$, so $w_{BC} + w_{AB}/\tau_1 = (1-2p_A) + 2p_A = 1$; likewise $1-\tau_1 = \frac{1/2-p_B}{p_A}$ gives $w_{CA}/(1-\tau_1) = 2p_A$. All six densities equal 1, so each marginal is Leb$[0,1]$, and the transport argument of Theorem 5.1 applies verbatim. Finally $\tau_i \in (0,1)$ for all i iff $\max_j p_j < \frac{1}{2}$ iff all three masses are positive, so every valid cyclic triple admits μ_p. □

Taking p the centroid recovers Theorem 5.1's bound ($\tau_i = \frac{1}{2}$, equal masses). For p off the centroid the directions are tilted non-median triples; e.g. $p = (\frac{9}{20}, \frac{3}{10}, \frac{1}{4})$ gives $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$ with masses $(\frac{1}{2}, \frac{1}{10}, \frac{2}{5})$. Theorem 6.3 thus proves the bound of Conjecture 1.1 for a two-parameter family of direction triples on the triangle, answering the finite-direction existence question of [5] affirmatively for the cyclic concurrent family. Figure 2 shows the running example.

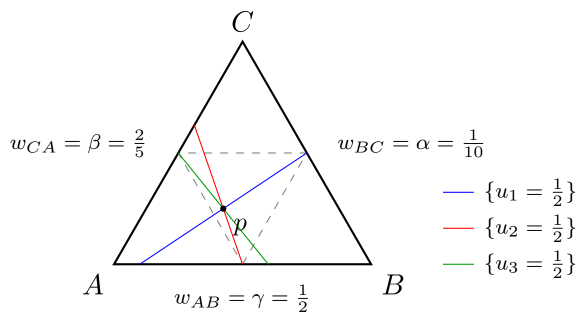

FIGURE 2. The weighted-perimeter witness of Theorem 6.3 at $p = (\frac{9}{20}, \frac{3}{10}, \frac{1}{4})$, i.e. $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$: the three mid-level lines $\{u_i = \frac{1}{2}\}$ concur at p, which lies in the open medial triangle (dashed), and μ_p puts the indicated masses uniformly on the edges.

Combining necessity (Proposition 6.1) with the explicit construction (Theorem 6.3) yields a *characterization*:

Corollary 6.4. *Call a direction triple* cyclic *if its vertex values are $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ with $\tau_i \in (0,1)$. A cyclic triple admits a uniform-marginal witness measure if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ concur. In that case the concurrence point p automatically lies in the open medial triangle ($p_j > 0$ and $p_j < \frac{1}{2}$ for all j), and the weighted-perimeter measure μ_p of Theorem 6.3 is a witness.*

Proof. The vertex-value matrix has $\det V = -(1 + \tau_1\tau_2\tau_3) \neq 0$, so Proposition 6.1 applies and gives necessity. Conversely, if the lines concur at p (a point of the affine plane of Δ, so $\sum_j p_j = 1$), then p is the unique solution of $Vp = \frac{1}{2}\mathbf{1}$, which in closed form reads $2(1 + \tau_1\tau_2\tau_3)\,p_A = 1 - \tau_3(1-\tau_2)$ and cyclically; each right-hand side is positive for $\tau_i \in (0,1)$, so $p_j > 0$, and the identities $\frac{1}{2} - p_C = \tau_1 p_A$, $\frac{1}{2} - p_A = \tau_3 p_B$, $\frac{1}{2} - p_B = \tau_2 p_C$ give $p_j < \frac{1}{2}$. Theorem 6.3 then supplies μ_p. □

Remark 6.5. The *anti-cyclic* pattern (all 0- and 1-roles exchanged) reduces to the cyclic one by a reflection of Δ (a transposition of two vertices), which maps the medial triangle to itself; Theorem 6.3, Corollary 6.4 and Theorem 6.6 below hold for it verbatim. All remaining role assignments are classified by Theorem 6.8 below.

The bound of Theorem 6.3 is in fact *attained* at every member of the family, and the tight covering is unique — a two-parameter family of sharp configurations extending the median rigidity of §5.

The coverage argument below rests on an elementary topological fact, which we isolate.

Pocket Lemma. Let H be an open convex subset of the plane of Δ such that $P := H \cap \operatorname{int}\Delta \neq \varnothing$ and $\overline{P} \cap \partial\Delta$ is finite. Then $H \subseteq \overline{\Delta}$; in particular H is bounded.

Proof. Suppose $y \in H \setminus \overline{\Delta}$, and choose a disk $B \subseteq P$. For $z \in B$ the segment $[z, y]$ lies in H by convexity and joins $\operatorname{int}\Delta$ to the complement of $\overline{\Delta}$; let $w(z)$ be the first point of $\partial\Delta$ on it, starting from z. Then $[z, w(z)) \subseteq H \cap \operatorname{int}\Delta$, so $w(z) \in \overline{P} \cap \partial\Delta$, a finite set. But for a fixed boundary point w, every z with $w(z) = w$ lies on the line through y and w, and finitely many lines cannot cover the disk B — a contradiction. □

Theorem 6.6 (tightness and rigidity of the cyclic family)**.** *Let p and u_1, u_2, u_3 be as in Theorem 6.3. Write $\alpha = 1 - 2p_A$, $\beta = 1 - 2p_B$, $\gamma = 1 - 2p_C$ (the edge masses of μ_p; $\alpha, \beta, \gamma > 0$, $\alpha + \beta + \gamma = 1$) and $S = \alpha\beta + \beta\gamma + \gamma\alpha$. Then the planks $P_i = \{u_i \in I_i\}$ with*

$$I_1 = \left[\tfrac{\alpha\gamma}{S},\, 1 - \tfrac{\alpha\beta}{S}\right], \qquad I_2 = \left[\tfrac{\beta\gamma}{S},\, 1 - \tfrac{\alpha\gamma}{S}\right], \qquad I_3 = \left[\tfrac{\alpha\beta}{S},\, 1 - \tfrac{\beta\gamma}{S}\right]$$

cover Δ, with $\sum \mathrm{rw} = \frac{\beta\gamma}{S} + \frac{\alpha\beta}{S} + \frac{\alpha\gamma}{S} = 1$. Moreover this is the unique covering of Δ with $\sum \mathrm{rw} = 1$ by three planks of positive width, one in each of the directions u_1, u_2, u_3. At p the centroid ($\alpha = \beta = \gamma = \frac{1}{3}$) it is $[\frac{1}{3}, \frac{2}{3}]^3$, recovering Theorem 5.1.

Proof. Write $l = (l_1, l_2, l_3) = (\alpha\gamma, \beta\gamma, \alpha\beta)/S$ and $h = (h_1, h_2, h_3) = \mathbf{1} - (\alpha\beta, \alpha\gamma, \beta\gamma)/S$, and note $\tau_1 = \frac{\gamma}{1-\alpha}$, $\tau_2 = \frac{\beta}{1-\gamma}$, $\tau_3 = \frac{\alpha}{1-\beta}$ from the formulas of Theorem 6.3. A direct substitution shows $(l, h) = (\lambda, \rho)$, the first solution of Proposition A.1; in particular $l_i < \tau_i < h_i$, and the traces of P_1, P_2, P_3 tile all three edges, so $\partial\Delta \subset \bigcup P_i$.

Step 1: an affine identity. Let $a_1 = \alpha(1-\alpha)$, $a_2 = \gamma(1-\gamma)$, $a_3 = \beta(1-\beta)$ and $\Lambda = a_1 u_1 + a_2 u_2 + a_3 u_3$. At the vertices: $\Lambda(A) = a_1\tau_1 + a_3 = \alpha\gamma + \beta(\alpha+\gamma) = S$, $\Lambda(B) = a_2 + a_3\tau_3 = \gamma(\alpha+\beta) + \alpha\beta = S$, $\Lambda(C) = a_1 + a_2\tau_2 = \alpha(\beta+\gamma) + \beta\gamma = S$. An affine function agreeing at three affinely independent points is constant: $\Lambda \equiv S$ on the plane of Δ. Moreover $\sum_i a_i = 1 - (\alpha^2 + \beta^2 + \gamma^2) = 2S$, and expanding (using $1 - \alpha = \beta + \gamma$ etc.)

$$\textstyle\sum_i a_i l_i = \frac{\alpha^2\gamma(\beta+\gamma) + \beta\gamma^2(\alpha+\beta) + \alpha\beta^2(\gamma+\alpha)}{S} = \frac{(\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2) + \alpha\beta\gamma}{S} = \frac{S^2 - \alpha\beta\gamma}{S},$$

since $S^2 = \sum \alpha^2\beta^2 + 2\alpha\beta\gamma$; the same expansion gives $\sum_i a_i(1 - h_i) = \frac{S^2 - \alpha\beta\gamma}{S}$, whence $\sum_i a_i h_i = 2S - \frac{S^2 - \alpha\beta\gamma}{S} = S + \frac{\alpha\beta\gamma}{S}$. Thus

$$\textstyle\sum_i a_i l_i = S - \frac{\alpha\beta\gamma}{S} < S < S + \frac{\alpha\beta\gamma}{S} = \sum_i a_i h_i. \tag{1}$$

Step 2: the planks cover Δ. Suppose not; the uncovered set U is relatively open (planks are closed) and misses $\partial\Delta$, and it decomposes as $U = \bigcup_\sigma P_\sigma$ over the eight sign patterns $\sigma \in \{\mathrm{lo}, \mathrm{hi}\}^3$, where $P_\sigma = H_\sigma \cap \operatorname{int}\Delta$ and H_σ is the open convex set $\{u_i < l_i\ (\sigma_i = \mathrm{lo}),\ u_i > h_i\ (\sigma_i = \mathrm{hi})\}$; write c_i for the active bound (l_i or h_i). We show each P_σ is empty.

First, $\overline{P_\sigma} \cap \partial\Delta$ is finite. Indeed, such a point x satisfies the closed pattern constraints ($u_j(x) \le l_j$ if $\sigma_j = \mathrm{lo}$, $u_j(x) \ge h_j$ if $\sigma_j = \mathrm{hi}$), and, lying on the tiled boundary, is covered by some P_j, so $u_j(x) \in [l_j, h_j]$; together these force $u_j(x) = c_j$ for that j. Each u_j is affine and nonconstant on each edge (its two vertex values there are distinct), so $\{u_j = c_j\}$ meets each edge in at most one point: at most $3 \times 3 = 9$ points in all. By the Pocket Lemma applied to H_σ (whose intersection with $\operatorname{int}\Delta$ is P_σ), $H_\sigma \subseteq \overline{\Delta}$ whenever $P_\sigma \neq \varnothing$; in particular H_σ is bounded.

No two of the lines $\ell_i = \{u_i = c_i\}$ are parallel: $u_i \parallel u_j$ would force $\tau_1(1 - \tau_2) = 1$, $\tau_2(1 - \tau_3) = 1$ or $\tau_3(1 - \tau_1) = 1$, impossible for $\tau \in (0, 1)^3$. If the three lines are concurrent, H_σ is an intersection of three open half-planes whose boundaries pass through one point, hence empty or an unbounded open cone; boundedness gives $H_\sigma = \varnothing$. Otherwise H_σ, if nonempty, is the open triangle with vertices $v_{ij} = \ell_i \cap \ell_j \in \overline{H_\sigma} \subseteq \overline{\Delta}$, and each v_{ij} satisfies the closed constraint in the third index k. Since $\Lambda \equiv S$, $u_k(v_{ij}) = (S - a_i c_i - a_j c_j)/a_k$, so that constraint reads $S \le \sum_m a_m c_m$ if $\sigma_k = \mathrm{lo}$, and $S \ge \sum_m a_m c_m$ if $\sigma_k = \mathrm{hi}$. For a *mixed* pattern both directions occur, forcing $S = \sum_m a_m c_m$; but then $u_k(v_{ij}) = c_k$, i.e. the three lines concur — excluded. For $\sigma = (\mathrm{lo}, \mathrm{lo}, \mathrm{lo})$ we get $S \le \sum a_i l_i$, and for $\sigma = (\mathrm{hi}, \mathrm{hi}, \mathrm{hi})$, $S \ge \sum a_i h_i$ — both contradict (1). Hence $U = \varnothing$: the planks cover, and $\sum \mathrm{rw} = 1$.

Step 3: uniqueness. Let (I'_i) be any covering with all $r'_i = |I'_i| > 0$ and $\sum r'_i = 1$. Since $u_{i\#}\mu_p = \mathrm{Leb}[0,1]$ (Theorem 6.3), $\mu_p(P'_i) = r'_i$, and $1 = \mu_p(\Delta) \le \sum \mu_p(P'_i) = 1$ forces the planks to partition μ_p a.e.; as μ_p has positive uniform density on each edge, the traces tile all three edges. By Proposition A.1, (I'_i) is $(\,[\lambda_i, \rho_i]\,) = (I_i)$, $(\,[0, \lambda_i]\,)$ or $(\,[\rho_i, 1]\,)$. The last two do not cover Δ: set $\delta = \frac{\alpha\beta\gamma}{2S^2}$ and let x^* be the unique point of the plane with $u_i(x^*) = \lambda_i + \delta$ for all i (consistent with $\Lambda \equiv S$ by (1) and $\sum a_i = 2S$; unique since V is invertible). Its barycentric coordinates are

$$x^* = \tfrac{1}{2S^2}\Big(\alpha\beta(1-\alpha)(1+\alpha-\beta),\ \beta\gamma(1-\beta)(\alpha+2\beta),\ \alpha\gamma(\alpha+\beta)(2-2\alpha-\beta)\Big),$$

all positive: $\alpha, \beta, \gamma > 0$ handle the leading factors (with $1-\alpha = \beta+\gamma > 0$), while $1+\alpha-\beta > 1-\beta > 0$, $\alpha+2\beta > 0$, and $2-2\alpha-\beta = \beta+2\gamma > 0$ using $\alpha+\beta+\gamma = 1$. So $x^* \in \operatorname{int}\Delta$ and $u_i(x^*) > \lambda_i$ for every i: x^* is uncovered by $(\,[0, \lambda_i]\,)$. Symmetrically the point y^* with $u_i(y^*) = \rho_i - \delta$, with coordinates $\frac{1}{2S^2}\big(\alpha\gamma(1-\alpha)(2\alpha+\beta), \alpha\beta(1-\beta)(1+\beta-\alpha), \beta\gamma(\alpha+\beta)(2-\alpha-2\beta)\big)$, positive likewise ($2\alpha+\beta > 0$, $1+\beta-\alpha > 1-\alpha > 0$, $2-\alpha-2\beta = \alpha+2\gamma > 0$), is uncovered by $(\,[\rho_i, 1]\,)$. Hence $(I'_i) = (I_i)$. □

Remark 6.7. At the centroid, $\delta = \frac{1}{6}$ and both witnesses degenerate to the point with $u_i \equiv \frac{1}{2}$ — the centroid itself, recovering the interior witness used in the proof of Theorem 5.1. The identities of Step 1 and the positivity of the witness coordinates, as well as exact coverage checks at several family points, are verified symbolically in the archived ancillary script `cyclic_family_tight_rigidity.py`.

Finally, the restriction to cyclic role assignments can be removed altogether. The plank problem is invariant under the *flip* of a normalized direction, $u \mapsto 1-u$, $I \mapsto 1-I$ (the planks, their widths, and uniformity of marginals are unchanged, and each mid-level line is fixed). Call the edge joining the 0- and 1-vertices of a direction its *full edge*. Flips reduce every role assignment to one of three shapes, and the existence question is settled in each:

Theorem 6.8 (three-direction characterization)**.** *Let u_1, u_2, u_3 be normalized directions on Δ, pairwise non-parallel. A probability measure μ on Δ with $u_{i\#}\mu = \mathrm{Leb}[0,1]$ for $i = 1,2,3$ exists if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ have a common point p. In that case $\sum \mathrm{rw} \ge 1$ for every covering of Δ by planks parallel to these directions, and exactly one of the following holds:*

(a) *the full edges are pairwise distinct: up to flips the triple is cyclic, p lies in the open medial triangle, and Theorems 6.3 and 6.6 apply verbatim (explicit witness μ_p; unique tight covering);*

(b) *all three directions have the same full edge e: then p is the midpoint of e, the uniform measure on e is a witness, and the bound follows from covering e alone (the trivial one-dimensional case recorded in Theorem 2.1).*

Proof. Necessity. If μ exists, its barycenter $p \in \Delta$ satisfies $u_i(p) = \mathbb{E}_\mu[u_i] = \frac{1}{2}$ for every i (no invertibility of V is needed), so the mid-lines concur at p.

Classification. (1) If the three full edges are distinct, flipping each direction so that its $(0,1)$-pair is oriented cyclically makes the triple cyclic; by Corollary 6.4, concurrence then places p in the open medial triangle and yields μ_p, and Theorem 6.6 the unique tight covering. (2) Suppose exactly two directions, say u_i and u_j, share their full edge e, and let x_v be the barycentric coordinate of the vertex v opposite e. After a flip, u_i and u_j agree on e, so $u_i - u_j = (\tau_i - \tau_j)x_v$ with $\tau_i \neq \tau_j$ (equality would make them parallel). Hence $\{u_i = \frac{1}{2}\} \cap \{u_j = \frac{1}{2}\} \subset \{x_v = 0\}$, the line through e, on which both mid-lines pass only through the midpoint of e; but the third direction is not full on e, its values at the endpoints of e being $\{0, \tau_k\}$ or $\{\tau_k, 1\}$, so its value at the midpoint is $\frac{\tau_k}{2}$ or $\frac{1+\tau_k}{2}$, never $\frac{1}{2}$. Thus the mid-lines of such a triple *never* concur; consistently, no witness measure exists ($\mathbb{E}_\mu[u_i] = \mathbb{E}_\mu[u_j] = \frac{1}{2}$ would force $\mathbb{E}_\mu[x_v] = 0$, hence $\operatorname{supp}\mu \subset e$, on which u_k is not onto $[0,1]$). This shape contributes to neither side of the equivalence. (3) If all three directions share the full edge e, then after flips each restricts to e as the identity parameter, so all mid-lines pass through the midpoint of e (concurrence always holds), and the uniform probability measure on e has all three pushforwards equal to $\mathrm{Leb}[0,1]$. For the bound, a covering of Δ covers e, on which the i-th plank traces an interval of length r_i; hence $\sum r_i \geq 1$. □

Remark 6.9. Theorem 6.8 answers the existence question of [5] completely for three directions on the triangle. In particular, *within the uniform-marginal witness-measure approach*, the remaining three-direction case is that of non-concurrent triples, for which no witness measure can exist. This is a statement about the reach of the method, not a reduction of Conjecture 1.1 on the triangle (which concerns arbitrarily many planks in arbitrary directions).

The concurrence condition is dimension-agnostic: the cyclic medians of Δ^d ($m_i(e_i) = \frac{1}{2}$, $m_i(e_{i+1}) = 0$, $m_i(e_{i-1}) = 1$, else $\frac{1}{2}$) have row-sums $\frac{d+1}{2}$, hence concur at the centroid in every dimension. (A caution specific to $d = 3$: there $m_{i+2} = 1 - m_i$, so this literal family collapses to two directions up to flips; this happens exactly when $d+1$ divides 4.) Section 8 constructs witness measures for a non-degenerate one-parameter family of concurrent quadruples on Δ^3, extending the transport method beyond the plane; for $d \geq 4$ the question is open (Problem 10.2).

7. The transport defect

Theorem 6.8 says the transport method certifies the full bound $\sum \mathrm{rw} \geq 1$ exactly when the mid-lines concur. We now quantify how the method degrades off the concurrent locus. Throughout, $u = (u_1, u_2, u_3)$ is a triple of normalized affine directions on Δ, pairwise non-parallel.

Definition 7.1. The *transport defect* of u is

$$D(u) \;=\; \inf\bigl\{D \geq 1 :\; \exists\, \mu \in \mathcal{P}(\Delta) \text{ with } u_{i\#}\mu \leq D \cdot \mathrm{Leb} \text{ on } [0,1] \text{ for } i = 1,2,3\bigr\}.$$

Proposition 7.2. *(i) The infimum is attained. (ii) Every covering of Δ by finitely many planks, each parallel to one of the three directions, satisfies $\sum \mathrm{rw} \geq 1/D(u)$. (iii) $D(u) = 1$ if and only if the mid-lines concur; otherwise $D(u) > 1$. (iv) $D(u) \leq 2$.*

Proof. (i) Take $D_n \downarrow D(u)$ and admissible μ_n; by weak-$*$ compactness of $\mathcal{P}(\Delta)$ pass to a limit μ. For every open interval J the set $u_i^{-1}(J)$ is relatively open in Δ, so $\mu(u_i^{-1}(J)) \leq \liminf_n \mu_n(u_i^{-1}(J)) \leq D(u)|J|$; open intervals determine the bound $u_{i\#}\mu \leq D(u)\,\mathrm{Leb}$ by outer regularity. (Taking $J = (-\delta, 1+\delta)$ also shows $D(u) \geq 1$.) (ii) For a plank P parallel to u_i with normalized interval I, $\mu(P \cap \Delta) = u_{i\#}\mu(I) \leq D(u)\,|I| \leq D(u)\,\mathrm{rw}(P)$; summing over a covering gives $1 \leq D(u) \sum \mathrm{rw}$. (iii) If $D(u) = 1$, the attained μ has $u_{i\#}\mu \leq \mathrm{Leb}$ with both sides of total mass 1 on $[0,1]$, hence $u_{i\#}\mu = \mathrm{Leb}$, and Theorem 6.8 forces concurrence; the converse is Theorem 6.8 again. (iv) The normalized uniform measure on Δ satisfies $\mu(u^{-1}(J)) \le 2|J|$ for every direction and interval, by the per-plank estimate in the proof of Theorem 2.1 with $d = 2$. ☐

The necessary condition of Proposition 6.1 can now be made quantitative. Define the *concurrence defect*

$$\delta(u) \;=\; \min_{p\in\Delta}\; \max_{i}\; \left|u_i(p) - \tfrac{1}{2}\right|, \tag{2}$$

a minimum of a continuous function on a compact set, hence attained; clearly $\delta(u) = 0$ if and only if the mid-lines concur. Moreover $\delta(u) < \frac{1}{2}$: at any interior p each $u_i(p) \in (0,1)$, since an affine function on Δ attaining its minimum 0 at an interior point would be identically 0.

Proposition 7.3. $D(u) \;\ge\; \dfrac{1}{1-2\delta(u)}$. *In particular, for the facet triple* $u = (\lambda_1, \lambda_2, \lambda_3)$ *one has* $\delta = \frac{1}{6}$ *and* $D \ge \frac{3}{2}$.

Proof. Let ν be a probability measure on $[0,1]$ with $\nu \le D\,\mathrm{Leb}$ and distribution function $F(t) \le \min(1, Dt)$. Its mean is $\int_0^1 (1-F) \ge \int_0^1 \max(0, 1-Dt)\,dt = \frac{1}{2D}$, and by the mirror argument the mean lies in $[\frac{1}{2D}, 1 - \frac{1}{2D}]$. If μ is admissible for D and p is its barycenter, then $p \in \Delta$ and the mean of $u_{i\#}\mu$ is $u_i(p)$, so $\delta(u) \le \max_i |u_i(p) - \frac{1}{2}| \le \frac{1}{2} - \frac{1}{2D}$, which rearranges to the claim. For the facets, $\sum_i \lambda_i(p) = 1$ forces some $\lambda_i(p) \le \frac{1}{3}$, so $\delta \ge \frac{1}{6}$; the centroid attains $\frac{1}{6}$. ☐

At the facets the moment bound is exact:

Theorem 7.4. *For the facet triple,* $D(\lambda_1, \lambda_2, \lambda_3) = \frac{3}{2}$. *The infimum is attained by the uniform probability measure on the perimeter of the inscribed triangle with vertices*

$$P_0 = (0, \tfrac{1}{3}, \tfrac{2}{3}), \qquad P_1 = (\tfrac{2}{3}, 0, \tfrac{1}{3}), \qquad P_2 = (\tfrac{1}{3}, \tfrac{2}{3}, 0),$$

whose three marginals all equal $\frac{3}{2} \cdot \mathbf{1}_{[0,2/3]}$.

Proof. The lower bound is Proposition 7.3. For the upper bound give each side of the loop mass $\frac{1}{3}$, uniformly in the parameter. The coordinate λ_1 is affine on each side, with values $0 \to \frac{2}{3}$ on P_0P_1 (density $\frac{1}{2}$ on $[0, \frac{2}{3}]$), $\frac{2}{3} \to \frac{1}{3}$ on P_1P_2 (density 1 on $[\frac{1}{3}, \frac{2}{3}]$), and $\frac{1}{3} \to 0$ on P_2P_0 (density 1 on $[0, \frac{1}{3}]$); the total is $\frac{3}{2}$ on $[0, \frac{2}{3}]$. The loop is invariant under the cyclic rotation of Δ, so the same holds for λ_2, λ_3. ☐

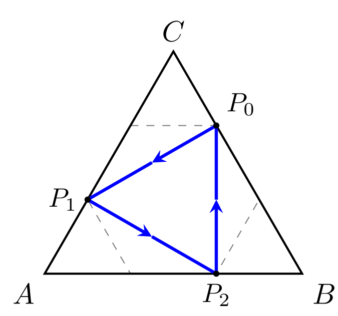

FIGURE 3. The witness of Theorem 7.4: the uniform measure on the inscribed loop $P_0P_1P_2$ (mass $\frac{1}{3}$ per side) has all three marginals equal to $\frac{3}{2} \cdot \mathbf{1}_{[0,2/3]}$, so $D = \frac{3}{2}$ for the facet triple. Dashed: the hexagon $\{\max_i \lambda_i \le \frac{2}{3}\}$, whose alternate vertices the loop connects.

Remark 7.5. Theorem 7.4 makes precise the complementarity of the two methods in this paper: for facet directions the tiling argument of Theorem 3.2 certifies the sharp bound 1, while the transport method cannot certify more than $1/D = \frac{2}{3}$ — and now provably certifies exactly that. Conversely, for the concurrent cyclic triples the transport bound is sharp (Theorems 6.3 and 6.6) where no facet tiling is available.

Near the concurrent locus the defect is small, with an explicit modulus; this gives bounds beyond the general constant of [4] on open sets of non-concurrent triples, for coverings in the corresponding direction families (see Remark 7.8).

Theorem 7.6. *Let $\tau^* \in (0,1)^3$ be a concurrent cyclic parameter with edge masses (α, β, γ) as in Theorem 6.3, and let $m = \min_i \min(\tau_i^*, 1-\tau_i^*)$. If $\tau \in (0,1)^3$ is a cyclic parameter with $\varepsilon := \max_i |\tau_i - \tau_i^*| < m$, then*

$$D(u_\tau) \;\le\; 1 + \frac{\varepsilon}{m(m-\varepsilon)}, \qquad \textit{hence} \qquad \sum \mathrm{rw} \;\ge\; 1 - \frac{\varepsilon}{m(m-\varepsilon)}$$

for every covering of Δ by planks parallel to the directions of τ.

Proof. Keep the *frozen* weights: let μ be the measure with masses (γ, α, β) on the edges (AB, BC, CA), uniform in the parameter, and push it forward by the *perturbed* directions. For u_1 (vertex values $(\tau_1, 0, 1)$) the marginal density is $\alpha + \gamma/\tau_1$ on $[0, \tau_1]$ and $\alpha + \beta/(1-\tau_1)$ on $[\tau_1, 1]$; at τ_1^* both equal 1 by the choice of weights in Theorem 6.3. Subtracting,

$$\Big(\alpha + \frac{\gamma}{\tau_1}\Big) - 1 = \frac{\gamma\,(\tau_1^* - \tau_1)}{\tau_1 \tau_1^*}, \qquad \Big(\alpha + \frac{\beta}{1-\tau_1}\Big) - 1 = \frac{\beta\,(\tau_1 - \tau_1^*)}{(1-\tau_1)(1-\tau_1^*)},$$

and since $\gamma, \beta \le 1$, $\tau_1^* \ge m$, $1-\tau_1^* \ge m$ and $\tau_1, 1-\tau_1 \ge m - \varepsilon$, both deviations are at most $\varepsilon/(m(m-\varepsilon))$ in absolute value; the same computation applies to u_2, u_3. Hence $u_{i\#}\mu \le (1+\varepsilon/(m(m-\varepsilon)))\,\mathrm{Leb}$ for all i, and Proposition 7.2(ii) with $1/(1+x) \ge 1-x$ gives the covering bound. ☐

Corollary 7.7. *Every covering of Δ by planks parallel to the directions of a cyclic parameter τ with $\max_i |\tau_i - \frac{1}{2}| \le \frac{1}{60}$ satisfies*

$$\sum \mathrm{rw} \;\ge\; \tfrac{27}{29} \;>\; 4\sqrt{3} - 6.$$

Proof. Apply Theorem 7.6 with $\tau^* = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ (the medians, concurrent with $\alpha = \beta = \gamma = \frac{1}{3}$), $m = \frac{1}{2}$, $\varepsilon = \frac{1}{60}$: the loss is $\frac{1/60}{(1/2)(1/2-1/60)} = \frac{2}{29}$. The comparison $\frac{27}{29} > 4\sqrt{3} - 6$ is $(\frac{201}{29})^2 = \frac{40401}{841} > 48$, i.e. $40401 > 40368$. ☐

Remark 7.8. The set of triples covered by Corollary 7.7 is a full-dimensional neighbourhood in the three-parameter space of cyclic triples, and the concurrent ones form a codimension-one subset of it: the corollary is a bound strictly inside non-concurrent territory, where by Theorem 6.8 no witness measure exists and $\sum \mathrm{rw} \ge 1$ remains open. We stress the honest comparison: the bound $4\sqrt{3} - 6$ of [4] is uniform over *all* direction triples, while ours improves on it only on this neighbourhood; far from the concurrent locus the moment bound (Proposition 7.3) shows the transport method degrades irreparably — at the facets down to $\frac{2}{3}$.

Remark 7.9. $D(u)$ is computable to certified precision by exact linear programming over skeleton measures. For example, for the (non-concurrent) cyclic parameter $\tau = (\frac{13}{25}, \frac{1}{2}, \frac{1}{2})$ one gets $\frac{225}{224} \le D(u_\tau) \le \frac{112}{111}$: the lower bound from Proposition 7.3 with $\delta = \frac{1}{450}$, the upper from the optimal edge-skeleton measure. Whether the moment bound is exact off the symmetric cases — as it is at the facets and, trivially, on the concurrent locus — we leave open; the ancillary script `transport_defect.py` contains the exact computations.

8. A witness family on the 3-simplex

Let $\Delta^3 = \mathrm{conv}(e_1, \dots, e_4)$ and fix $\sigma \in (0,1)$. Define four normalized affine directions by their vertex values (indices mod 4):

$$u_i(e_i) = 0, \qquad u_i(e_{i+1}) = 1, \qquad u_i(e_{i+2}) = \sigma, \qquad u_i(e_{i+3}) = 1-\sigma. \tag{3}$$

Each row sums to 2, so the mid-hyperplanes $\{u_i = \frac{1}{2}\}$ concur at the centroid (Proposition 6.1); at $\sigma = \frac{1}{2}$ the $\{u_i = \frac{1}{2}\}$ are the "median" planes through an edge and the midpoint of the opposite edge. The four directions are pairwise non-parallel for every σ: an affine relation $u_{i+1} = c\,u_i + e$ forces, comparing vertex values, $(1-\sigma)^2 = 1$, and $u_{i+2} = c\,u_i + e$ forces $2\sigma(1-\sigma) = 0$, both impossible on $(0,1)$.

Theorem 8.1. *Let $\sigma \in (0, \frac{1}{2}]$ and set*

$$a(\sigma) = \frac{(1-\sigma)(1-2\sigma)}{2(\sigma^2 - 4\sigma + 2)}, \qquad b(\sigma) = \frac{\sigma(2-3\sigma)}{2(\sigma^2 - 4\sigma + 2)}.$$

The measure μ_σ placing mass $a(\sigma)$ on each of the four cycle edges e_ie_{i+1} and mass $b(\sigma)$ on each of the two diagonals e_1e_3, e_2e_4, uniformly in the parameter on each edge, is a probability measure on the 1-skeleton of Δ^3 with

$$u_{i\#}\mu_\sigma = \mathrm{Leb}[0,1] \qquad \textit{for } i = 1, \dots, 4.$$

Consequently every covering of Δ^3 by planks parallel to these four directions satisfies $\sum \mathrm{rw} \geq 1$. For $\sigma \in (\frac{1}{2}, 1)$ the same holds with the flip $u \mapsto 1-u$ (which exchanges $\sigma \leftrightarrow 1-\sigma$ and reverses the cyclic order). Moreover μ_σ is the unique cyclically invariant edge-uniform witness, and at $\sigma = \frac{1}{2}$ (where $a = 0$, $b = \frac{1}{2}$) it is the unique witness supported on the 1-skeleton at all.

Proof. Positivity of the weights on $(0, \frac{1}{2}]$: the roots of $\sigma^2 - 4\sigma + 2$ are $2 \pm \sqrt{2}$, so the denominator is positive on $(0, \frac{1}{2}] \subset (0, 2-\sqrt{2})$, and the numerators are nonnegative there (vanishing only at $\sigma = \frac{1}{2}$, where the middle piece below is empty).

By the cyclic symmetry of (3) it suffices to compute the u_1-marginal. The six edges push forward as follows ($\sigma < \frac{1}{2}$; each edge uniform, mass as assigned): e_1e_2, values $0 \to 1$, density a on $[0,1]$; e_2e_3, values $1 \to \sigma$, density $a/(1-\sigma)$ on $[\sigma, 1]$; e_3e_4, values $\sigma \to 1-\sigma$, density $a/(1-2\sigma)$ on $[\sigma, 1-\sigma]$; e_4e_1, values $1-\sigma \to 0$, density $a/(1-\sigma)$ on $[0, 1-\sigma]$; e_1e_3, values $0 \to \sigma$, density b/σ on $[0,\sigma]$; e_2e_4, values $1 \to 1-\sigma$, density b/σ on $[1-\sigma, 1]$. The total density is

$$a + \frac{a}{1-\sigma} + \frac{b}{\sigma} \ \text{ on } [0,\sigma] \text{ and on } [1-\sigma, 1], \qquad a + \frac{2a}{1-\sigma} + \frac{a}{1-2\sigma} \ \text{ on } [\sigma, 1-\sigma],$$

and a direct expansion shows both expressions equal 1 exactly for the stated $a(\sigma), b(\sigma)$; the total mass $4a + 2b = 1$ is then automatic, since pushforward preserves mass. The covering bound follows as in Proposition 7.2(ii) with $D = 1$. Uniqueness among cyclically invariant edge-uniform measures: the middle-piece equation determines a and either outer equation then determines b. (Exact linear algebra, recorded in the ancillary files, shows uniqueness among *all* edge-uniform weightings.) At $\sigma = \frac{1}{2}$ the values (3) make u_i *constant* ($\equiv \frac{1}{2}$) on the cycle edge $e_{i+2}e_{i+3}$; any skeleton measure giving positive mass to a cycle edge would push forward to an atom at $\frac{1}{2}$ under the corresponding direction, so all four cycle-edge masses vanish, and on the two diagonals each direction is affine onto $[0, \frac{1}{2}]$ or $[\frac{1}{2}, 1]$, forcing the uniform density $\frac{1}{2} + \frac{1}{2}$ split. □

Proposition 8.2. *For $d \geq 4$, consider the cyclic medians of Δ^d: $u_i(e_i) = 0$, $u_i(e_{i+1}) = 1$ and $u_i(e_j) = \frac{1}{2}$ otherwise (indices mod $d+1$). Then* every *edge e_je_k of Δ^d is constant for some direction, and consequently no measure supported on the 1-skeleton has uniform marginals for all $d+1$ directions.*

Proof. Fix an edge $\{j,k\}$. Removing j,k from the cycle on $d+1 \geq 5$ vertices leaves at least three vertices in at most two arcs, so some arc contains two cyclically consecutive vertices $\{m, m+1\}$ disjoint from $\{j,k\}$; then u_m takes the value $\frac{1}{2}$ at both e_j and e_k, hence is constant on the edge. A skeleton measure with positive mass on e_je_k pushes forward under u_m to a measure with an atom at $\frac{1}{2}$, which is not $\leq D \cdot \mathrm{Leb}$ for any finite D; so all edge masses vanish. □

Remark 8.3. The obstruction is specific to the all-$\frac{1}{2}$ interior values; whether distinct interior values (the $d \geq 4$ analogue of (3)) revive the skeleton, or the mass must move to 2-faces, is open and part of Problem 10.2.

9. A normalizer obstruction

For a direction u let $\ell_K(u)$ be the length of the longest chord of K parallel to u; always $\ell_K(u) \leq \mathrm{w}_K(u)$. Bakaev–Yehudayoff's chord bound $\sum_i \mathrm{w}(P_i)/\ell_K(u_i) \geq 1$ [4] follows from Bang's sign argument via the chord lemma. Consider interpolating the normalizer, $N_c = (1-c)\ell + c\,\mathrm{w}$, $c \in [0,1]$: $N_c = \ell$ at $c=0$ (chord bound) and $N_c = \mathrm{w}$ at $c = 1$ (Conjecture 1.1).

Proposition 9.1. *For a convex body K, a direction u with $\|u\| = a/2$ and $\ell = \ell_K(u) \geq a$, the body $(K-u) \cap (K+u)$ contains a homothet of K of ratio $(\ell - a)/\ell$* [6, Lem. 2.3]*, and this ratio is sharp: for K a segment of length ℓ parallel to u, the set $(K-u)\cap(K+u)$ is a segment of length exactly $\ell - a$, and no homothet of K of larger ratio fits in it.*

Proof. The containment is [6, Lem. 2.3]. For sharpness let $K = [y,z]$ with $z - y = \ell\,\hat{u}$ and $u = \frac{a}{2}\hat{u}$: then $K - u = [y - \frac{a}{2}\hat{u},\, z - \frac{a}{2}\hat{u}]$ and $K + u = [y + \frac{a}{2}\hat{u},\, z + \frac{a}{2}\hat{u}]$, whose intersection is $[y + \frac{a}{2}\hat{u},\, z - \frac{a}{2}\hat{u}]$, a segment of length $\ell - a$; a homothet of K contained in it has ratio at most $(\ell - a)/\ell$. □

Remark 9.2 (methodological obstruction). The reading of Proposition 9.1 that matters here is about the *sign argument*: iterating the chord lemma over the directions u_j places a Bang-system witness $x_0 + \sum_j \pm u_j$ as soon as $\sum_j \mathrm{w}(P_j)/\ell_j < 1$, which is how the chord bound $\sum \mathrm{w}(P_j)/\ell_j \geq 1$ of [4] follows. The movement budget in direction u is governed by the homothety ratio of Proposition 9.1, which is exactly $(\ell - a)/\ell$ and cannot be stretched: the placement route proves the bound for the normalizer $N = \ell$ and for no direction-normalizer $N > \ell$; in particular it does not establish $\sum_i \mathrm{w}(P_i)/N_c(u_i) \geq 1$ for any $c > 0$. This does *not* show the latter inequality false; it shows this route cannot prove it. Beating $2/(1+\sqrt{d})$ by such a normalizer therefore requires a witness that is not a Bang-system point, i.e. one exploiting the coupling of the covering — consistent with the sharpness of [4] for the cube.

10. Open problems

Problem 10.1. The three-plank case for non-concurrent tilted triples on the triangle: Theorem 6.8 settles the concurrent case, and shows that for non-concurrent triples no witness measure exists, so a different mechanism is needed there. (This is the three-plank sub-problem; Conjecture 1.1 for the triangle concerns coverings by arbitrarily many planks and is not reduced to it.)

Problem 10.2. Theorem 6.8 settles the existence of uniform-marginal witness measures for three directions on the triangle, and Theorem 8.1 gives an affirmative one-parameter family of concurrent quadruples on Δ^3. For $d \geq 3$ the concurrence condition of Proposition 6.1 remains necessary; is it sufficient for $d+1$ directions on Δ^d? For the cyclic medians of Δ^d with $d \geq 4$, any witness must vanish on the 1-skeleton (Proposition 8.2); the first open cases are supports on 2-faces, or interior vertex values as in (3).

Problem 10.3. Is the moment bound of Proposition 7.3 always exact, i.e. does $D(u) = 1/(1 - 2\delta(u))$ hold for every pairwise non-parallel triple on the triangle (as it does at the facets and on the concurrent locus)? In particular, is $\sup_u D(u) = \frac{3}{2}$, attained at the facet triple? A small exact sweep (ancillary files) finds no triple with $\delta(u) > \frac{1}{6}$.

Problem 10.4. Whether $\sum \mathrm{w}(P_i)/N_c(u_i) \geq 1$ holds for some $c > 0$ (a coupling certificate), which would improve the general bound past $2/(1 + \sqrt{d})$.

Appendix A. Edge tilings of cyclic triples

Throughout, $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ is a cyclic triple with arbitrary $\tau_1, \tau_2, \tau_3 \in (0, 1)$ (*no* concurrence is assumed), and $I_i = [l_i, h_i] \subset [0, 1]$, $l_i < h_i$, are three intervals. Say (I_1, I_2, I_3) satisfies the *edge-tiling conditions* if on each edge of Δ the traces $T_i = \{x \in \text{edge} : u_i(x) \in I_i\}$ cover the edge and have pairwise disjoint interiors. Indices are mod 3.

Proposition A.1. *For every $\tau \in (0,1)^3$ the edge-tiling conditions have exactly three solutions: with*

$$\lambda_i = \frac{\tau_i\big(1 - \tau_{i+1}(1 - \tau_{i+2})\big)}{1 + \tau_1\tau_2\tau_3}, \qquad \rho_i = \frac{1 - \tau_{i+2}(1 - \tau_i)}{1 + (1 - \tau_1)(1 - \tau_2)(1 - \tau_3)},$$

they are $\big([\lambda_i, \rho_i]\big)_i$, $\big([0, \lambda_i]\big)_i$, and $\big([\rho_i, 1]\big)_i$. One has $0 < \lambda_i < \tau_i < \rho_i < 1$ for all i and all τ. For $\tau_i \equiv \frac{1}{2}$ (the medians) the solutions are $[\frac{1}{3}, \frac{2}{3}]^3$, $[0, \frac{1}{3}]^3$, $[\frac{2}{3}, 1]^3$.

Setup. Parametrize the edges by $t \in [0, 1]$ in the orientations $A \to B$, $B \to C$, $C \to A$. From the vertex values, each edge sees one direction with values $[0, \tau_c]$ (the *low* direction c, $u_c(t) = \tau_c(1 - t)$), one with values $[0, 1]$ (the *full* direction f, $u_f(t) = t$), and one with values $[\tau_g, 1]$ (the *high* direction g, $u_g(t) = 1 - (1 - \tau_g)t$), with roles

$$(c, f, g) = (1, 2, 3) \text{ on } AB, \qquad (3, 1, 2) \text{ on } BC, \qquad (2, 3, 1) \text{ on } CA.$$

Inverting, the traces on an edge with roles (c, f, g) are

$$\begin{aligned} &T_c = \Big[\max\big(0, 1 - \tfrac{h_c}{\tau_c}\big),\, 1 - \tfrac{l_c}{\tau_c}\Big] \ \text{(nondegenerate iff } l_c < \tau_c), \qquad T_f = [l_f, h_f], \\ &T_g = \Big[\tfrac{1 - h_g}{1 - \tau_g},\, \min\big(1, \tfrac{1 - l_g}{1 - \tau_g}\big)\Big] \ \text{(nondegenerate iff } h_g > \tau_g). \end{aligned} \tag{4}$$

Classify each I_i by its position relative to τ_i: type L if $h_i \leq \tau_i$, type R if $l_i \geq \tau_i$, type M if $l_i < \tau_i < h_i$; every I_i is of exactly one type. Two elementary observations, used throughout:

(O1) If I_c is of type M then $T_c = [0,\, 1 - l_c/\tau_c]$ (it contains the left endpoint); if I_c is of type R then T_c is degenerate (a point or empty). Dually, if I_g is of type M then $T_g = [(1 - h_g)/(1 - \tau_g),\, 1]$; if I_g is of type L then T_g is degenerate.

(O2) The tiling condition is insensitive to degenerate traces: a degenerate trace covers at most one point, and since the nondegenerate traces are finitely many closed sets whose union contains $[0, 1]$ minus finitely many points, that union is all of $[0, 1]$. So on each edge the *nondegenerate* traces tile $[0, 1]$; in particular, finitely many closed intervals of positive length with disjoint interiors covering $[0, 1]$ abut consecutively from 0 to 1. Consequently the case analysis below requires no genericity in τ or in the endpoints: whenever an endpoint coincidence (e.g. $h_i = \tau_i$ or $l_i = \tau_i$) makes a trace degenerate, (O2) discards it, and every case argument uses only the defining non-strict inequalities of the types.

Symmetries. Two operations act on the data $(\tau; I_1, I_2, I_3)$ and transport edge-tilings to edge-tilings.

The *rotation* ϱ is induced by the affine map $A \to B \to C \to A$ of Δ: it carries the cyclic triple with parameter (τ_1, τ_2, τ_3) to the one with parameter (τ_2, τ_3, τ_1) and acts by $(I_1, I_2, I_3) \mapsto (I_2, I_3, I_1)$, shifting type vectors cyclically.

The *reflected flip* ς is the affine reflection of Δ swapping B and C, composed with the flip $s \mapsto 1-s$ of each normalized direction. The flip replaces each u_i by $1-u_i$ and each I_i by $1-I_i$ ($1-[l,h] = [1-h, 1-l]$), leaving every plank, every width, every trace and the tiling conditions unchanged; the reflection then restores the cyclic normalization (value 0 and 1 at the prescribed vertices), and the composite carries the parameter (τ_1, τ_2, τ_3) to $(1-\tau_1,\ 1-\tau_3,\ 1-\tau_2)$, acting by $(I_1, I_2, I_3) \mapsto (1-I_1,\ 1-I_3,\ 1-I_2)$; on type vectors it acts by $(X_1, X_2, X_3) \mapsto (\bar{X}_1, \bar{X}_3, \bar{X}_2)$, where the bar exchanges $\mathrm{L} \leftrightarrow \mathrm{R}$ and fixes M.

One checks $\varsigma^2 = \mathrm{id}$ and $\varsigma\varrho\varsigma = \varrho^{-1}$, so $G = \langle \varrho, \varsigma \rangle$ is a group of order 6 (isomorphic to S_3). Each $g \in G$ maps the edge-tiling problem with parameter τ *bijectively* onto the problem with parameter $g(\tau)$. Since Proposition A.1 is proved for *all* $\tau \in (0,1)^3$ simultaneously, it suffices to treat one representative of each G-orbit of the 27 type vectors $(X_1, X_2, X_3) \in \{\mathrm{L}, \mathrm{M}, \mathrm{R}\}^3$:

$$\mathrm{MMM};\quad \mathrm{LLL}\ (\sim \mathrm{RRR});\quad \mathrm{MML};\quad \mathrm{LLM};\quad \mathrm{LLR};\quad \mathrm{LMR};\quad \mathrm{LRM}$$

(orbit sizes $1, 2, 6, 6, 6, 3, 3$, summing to 27).

Case MMM. By (O1), on every edge $T_c = [0,\ 1 - l_c/\tau_c]$ and $T_g = [(1-h_g)/(1-\tau_g),\ 1]$, both nondegenerate, and T_f has positive length. Disjointness of interiors forces T_f inside the gap between them, and covering forces the abutments

$$l_f = 1 - \frac{l_c}{\tau_c}, \qquad h_f = \frac{1-h_g}{1-\tau_g}.$$

Reading this on the three edges gives two decoupled cyclic systems,

$$l_2 = 1 - \frac{l_1}{\tau_1}, \quad l_1 = 1 - \frac{l_3}{\tau_3}, \quad l_3 = 1 - \frac{l_2}{\tau_2}; \qquad h_1 = \frac{1-h_2}{1-\tau_2}, \quad h_2 = \frac{1-h_3}{1-\tau_3}, \quad h_3 = \frac{1-h_1}{1-\tau_1}.$$

Each system composes three decreasing affine maps, so the composite self-map (of l_1, resp. h_1) is affine with negative slope ($-1/(\tau_1\tau_2\tau_3)$, resp. $-1/\prod(1-\tau_i)$), hence has a unique fixed point: the solutions are $l_i = \lambda_i$, $h_i = \rho_i$ (direct substitution verifies the formulas; e.g. $1 - \lambda_1/\tau_1 = \big(\tau_1\tau_2\tau_3 + \tau_2(1-\tau_3)\big)/(1+\tau_1\tau_2\tau_3) = \lambda_2$). The chain $0 < \lambda_i < \tau_i < \rho_i < 1$ reduces to $\tau_{i+1}(\tau_{i+2}-1) < \tau_1\tau_2\tau_3$ and $\tau_i(1-\tau_{i+1}) < 1$, both automatic; so the solution is of type MMM and, by construction, tiles each edge as $[0, l_f] \cup [l_f, h_f] \cup [h_f, 1]$ (Figure 4).

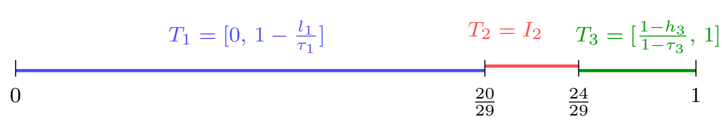

FIGURE 4. The MMM edge tiling on AB (roles $(c,f,g) = (1,2,3)$) for the running example $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$: the traces of $I = ([\frac{5}{29}, \frac{25}{29}], [\frac{20}{29}, \frac{24}{29}], [\frac{4}{29}, \frac{9}{29}])$ abut at $\frac{20}{29}$ and $\frac{24}{29}$ and tile $[0,1]$, as forced in Case MMM.

Case LLL. All T_g are degenerate, so each edge is tiled by $T_c = [1 - h_c/\tau_c,\ 1 - l_c/\tau_c]$ (the full image, as $I_c \subset [0, \tau_c]$) and $T_f = I_f$ with $h_f \le \tau_f < 1$. The tile containing 1 must be T_c, forcing $l_c = 0$. If $h_c = \tau_c$ then $T_c = [0, 1]$ and T_f has empty interior, impossible; so T_f contains 0 ($l_f = 0$) and abuts T_c: $h_f = 1 - h_c/\tau_c$. On the three edges: $l_i = 0$ for all i, and the h_i satisfy the *same* cyclic system as the l_i in case MMM, so $h_i = \lambda_i$, giving $\big([0, \lambda_i]\big)_i$ (type L since $\lambda_i < \tau_i$). Applying the reflection symmetry, the orbit representative RRR yields the unique solution $h_i = 1$, $l_i = \rho_i$, i.e. $\big([\rho_i, 1]\big)_i$; equivalently, the l_i satisfy the same cyclic system as the h_i in case MMM.

Case MML (impossible). On AB: T_3 is degenerate (I_3 of type L), so $[0, 1]$ is tiled by $T_1 = [0,\ 1 - l_1/\tau_1]$ (O1) and $T_2 = I_2$. If $1 \in T_1$ then $T_1 = [0, 1]$ and T_2 has empty interior, impossible; so $h_2 = 1$. Then on BC the high trace (4) is $T_2 = [\frac{1-h_2}{1-\tau_2}, \min(1, \frac{1-l_2}{1-\tau_2})] = [0, 1]$ (as $h_2 = 1$ and $l_2 < \tau_2$), leaving the full trace $T_1 = I_1$ (positive length) with empty interior — contradiction.

Case LLM (impossible). On BC ($(c, f, g) = (3, 1, 2)$): the high trace is degenerate (I_2 of type L), so $[0, 1]$ is tiled by $T_3 = [0,\ 1 - l_3/\tau_3]$ (O1, I_3 of type M) and $T_1 = I_1$ with $h_1 \le \tau_1 < 1$. The tile containing 1 must be T_3, so $l_3 = 0$ and $T_3 = [0, 1]$, leaving T_1 with empty interior — contradiction.

Case LLR (impossible). On BC both compressed traces are degenerate (I_3 of type R gives a degenerate low trace, I_2 of type L a degenerate high trace), so $T_1 = I_1$ alone tiles $[0, 1]$, forcing $h_1 = 1 > \tau_1$ — contradicting type L.

Case LMR (impossible). *On BC* ($(c, f, g) = (3, 1, 2)$): the low trace is degenerate (I_3 of type R), so $T_1 = I_1$ and $T_2 = [\frac{1-h_2}{1-\tau_2},\ 1]$ (O1, I_2 of type M) tile $[0, 1]$. If $T_2 = [0, 1]$ then T_1 has empty interior, impossible; so T_1 contains 0 and abuts T_2: $l_1 = 0$ and $h_1 = \frac{1-h_2}{1-\tau_2}$.

On CA ($(c, f, g) = (2, 3, 1)$): the high trace is degenerate (I_1 of type L), so $T_2 = [0,\ 1 - l_2/\tau_2]$ (O1) and $T_3 = I_3$ tile $[0, 1]$. If $T_2 = [0, 1]$ then T_3 has empty interior, impossible; so $h_3 = 1$ and $l_3 = 1 - l_2/\tau_2$. Since I_3 has positive width, $l_3 < 1$, which forces $l_2 > 0$.

On AB ($(c, f, g) = (1, 2, 3)$): as $h_3 = 1$, the high trace is $T_3 = \big[0, \min\big(1, \frac{1-l_3}{1-\tau_3}\big)\big]$. Two subcases. If $\frac{1-l_3}{1-\tau_3} \ge 1$, then $T_3 = [0, 1]$ and the full trace $T_2 = I_2$ (positive length) has empty interior — contradiction. Otherwise $T_3 = \big[0,\ \frac{1-l_3}{1-\tau_3}\big] = \big[0,\ \frac{l_2}{\tau_2(1-\tau_3)}\big]$; since $\tau_2(1 - \tau_3) < 1$ and $l_2 > 0$, its right endpoint strictly exceeds l_2, so the interiors of T_3 and of $T_2 = [l_2, h_2]$ overlap on an interval just above l_2 — contradiction.

Case LRM (impossible). On CA: both compressed traces are degenerate (I_2 of type R, I_1 of type L), so $T_3 = I_3$ alone tiles $[0, 1]$, i.e. $I_3 = [0, 1]$. Then on AB the high trace (4) is $T_3 = [\frac{1-h_3}{1-\tau_3}, \min(1, \frac{1-l_3}{1-\tau_3})] = [0, 1]$, leaving $T_2 = I_2$ (positive length) with empty interior — contradiction.

All orbits are exhausted, proving Proposition A.1. □

The three solutions have also been verified by exact rational searches at $\tau = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $\tau = (\frac{1}{2}, \frac{2}{3}, \frac{1}{3})$, and the median enumeration was verified by two further independent exact computations; all scripts are archived as ancillary files.

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